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COMBINED COOLING LOOP WITHIN ELECTROCHEMICAL PLANT

Abstract

The present disclosure advantageously provides an improved cooling system for an electrochemical plant. The configurations disclosed herein provide advantages and improvements in a cooling system for the electrochemical plant. The cooling system advantageously cools multiple subsystems within the plant using one unified cooling loop, thereby easing maintenance and access to various components within the plant, minimizing or reducing the amount of process piping within the plant used to cool the multiple subsystems, and reducing the complexity of the overall plant.

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Background/Summary

[0001] The present patent document claims the benefit of U.S. Provisional Patent Application No. 63/556,790, filed Feb. 22, 2024, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The following disclosure relates to an electrochemical plant, and in particular for a high-capacity electrochemical plant having a combined cooling loop to cool various subsystems within the plant.

BACKGROUND

[0003] Electrolyzer systems use electrical energy to drive a chemical reaction. For example, water is split to form hydrogen and oxygen. The products may be used as energy sources for later use. In recent years, improvements in operational efficiency have made electrolyzer systems competitive market solutions for energy storage, generation, and/or transport. For example, the cost of generation may be below \$10 per kilogram of hydrogen in some cases. Increases in efficiency and/or improvements in operation will continue to drive installation of electrolyzer systems.

[0004] Conventional electrolysis plants are commonly designed to function within power levels ranging from 1 to 20 megawatts (MW), necessitating effective cooling to ensure smooth plant operation. Cooling is imperative for various subsystems in an electrolysis plant, encompassing the anode loop process water, the electrical system, and, if applicable, the cathode loop process water. Traditionally, these subsystems operate on distinct cooling loops, leading to complexity and additional expenses.

SUMMARY

[0005] In one embodiment, a cooling system for an electrochemical plant is provided. The cooling system includes a cooling module having a plurality of dry coolers or wet coolers configured to transfer coolant and reject waste heat generated in a plurality of separate modules of the electrochemical plant. The cooling system further includes a cooling loop having cooling lines configured to provide fluid communication between the cooling module and each module of the plurality of separate modules within the electrochemical plant. The cooling module is configured to transfer the coolant via the cooling loop to each module of the plurality of separate modules.

Additionally, the cooling module is configured to receive, via the cooling loop, the coolant from each module of the plurality of separate modules and reject the waste heat collected by the coolant received from each module of the plurality of separate modules to a surrounding environment.

[0006] In another embodiment, a method for cooling an electrochemical plant using a cooling system is provided. The method includes transferring coolant via a cooling loop of the cooling system, by a cooling module of the cooling system, to each module of a plurality of separate modules. The method also includes: receiving coolant, via the cooling loop of the cooling system, from each module of the plurality of separate modules; and rejecting, by the cooling module, waste heat collected by the coolant received from each module of the plurality of separate modules to a surrounding environment. The cooling module includes a plurality of dry coolers or wet coolers configured to transfer coolant and reject waste heat generated in the plurality of separate modules of the electrochemical plant. The cooling loop includes cooling lines configured to provide fluid communication between the cooling module and each module of the plurality of separate modules within the electrochemical plant.

[0007] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key

features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Exemplary embodiments are described herein with reference to the following drawings.

[0009] FIG. 1A depicts an example of an electrochemical or electrolytic cell.

[0010] FIG. 1B depicts an example of a system including an electrochemical stack having a plurality of electrochemical cells of FIG. 1A.

[0011] FIG. 2 depicts an example of an electrochemical plant.

[0012] FIG. 3 depicts an additional example of an electrochemical plant.

[0013] FIGS. 4 and 5 depict examples of a cooling system of an electrochemical plant.

[0014] FIG. 6 depicts a flow chart explaining a method of cooling an electrochemical plant using a cooling system.

[0015] FIG. 7 depicts an example communication system between a cooling system and a computing device having a controller over a connected network.

[0016] FIG. 8 depicts an example of a computing device having a controller.

DETAILED DESCRIPTION

[0017] The present disclosure advantageously provides an improved electrochemical plant operating multiple subsystems on one unified cooling loop to efficiently cool all modules or subsystems of the plant. The configurations disclosed herein provide advantages and improvements in a cooling system for the electrochemical plant. The cooling system advantageously cools multiple subsystems within the plant using one unified cooling loop thereby easing maintenance and access to various components within the plant, minimizing or reducing the amount of process piping within the plant used to cool the multiple subsystems, and reducing the complexity of the overall plant.

Electrochemical Cells/Stacks

[0018] FIG. 1A depicts an example of an electrochemical cell for the production of hydrogen gas and oxygen gas through the splitting of water. The electrochemical cell includes a cathode, an anode, and a membrane positioned between the cathode and anode. Within the water-splitting electrolysis reaction, one interface runs an oxygen evolution reaction (OER) while the other interface runs a hydrogen evolution reaction (HER). For example, the anode reaction is $\text{H}_2\text{O} \rightarrow \frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ and the cathode reaction is $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$. The water electrolysis reaction has recently assumed great importance and renewed attention as a potential foundation for a decarbonized “hydrogen economy.”

[0019] FIG. 1B depicts an example of a system including an electrochemical stack having a plurality of electrochemical cells of FIG. 1A. In certain examples, the electrochemical stacks may contain 50-1000 cells, 50-100 cells, 500-700 cells, or more than 1000 cells. Any number of cells may make up a stack. The electrochemical cells within the electrochemical stack may be configured to operate with 200 mV or less of pure resistive loss when operating at a high current density (e.g., at least 3 Amps/cm², at least 4 Amps/cm², at least 5 Amps/cm², at least 6 Amps/cm², at least 7 Amps/cm², at least 8 Amps/cm², at least 9 Amps/cm², at least 10 Amps/cm², at least 11 Amps/cm², at least 12 Amps/cm², at least 13 Amps/cm², at least 14 Amps/cm², at least 15 Amps/cm², at least 16 Amps/cm², at least 17 Amps/cm², at least 18 Amps/cm², at least 19 Amps/cm², at least 20 Amps/cm², at least 25 Amps/cm², at least 30 Amps/cm², in a range of 1-30 Amps/cm², in a range of 3-20 Amps/cm², in a range of 3-15 Amps/cm², in a range

of 3-10 Amps/cm.sub.2, or in a range of 10-20 Amps/cm.sup.2). In additional examples, the amount of water (e.g., deionized (DI) water) transferred to or circulated through each cell of the stack may be in a range of 0.25-5 mL/Amp/cell/min.

[0020] As illustrated in the system of FIG. 1B, water (H.sub.2O) may be supplied to the anodic inlet of an electrolytic cell stack 12. In some embodiments, only the anodic inlet of the cell stack 12 may receive water. In these embodiments, the cathode side of the cell stack 12 may not receive water (e.g., a dry cathode side may be used). In another embodiment, a cathode inlet may also receive water, wherein the water may be supplied to the cathode Inlet to cool the cell stack 12 during electrolysis.

[0021] The water supplied to the anodic inlet flows to an anodic inlet manifold that distributes the water to the anode side of the plurality of cells contained with the cell stack 12. In embodiments where water is supplied to the cathode inlet, water supplied to the cathode inlet flows to a cathodic inlet manifold that distributes the water to the cathode side of the plurality of cells in the cell stack 12.

[0022] During electrolysis, oxygen (O.sub.2) is produced at the anode side of the electrolytic cells and hydrogen (H.sub.2) is produced at the cathode side of the electrolytic cells. Specifically, a water splitting electrolysis reaction is configured to take place within each individual cell in the cell stack 12. Each cell includes one interface (the anode side of the cell) configured to run an oxygen evolution reaction (OER) and another interface (the cathode side of the cell) configured to run a hydrogen evolution reaction (HER), such as depicted in FIG. 1A.

[0023] During electrolysis, some of the water supplied to the anode side of an electrolytic cell may not be converted into oxygen. Accordingly, a two-phase flow of oxygen and unreacted water is outlet from each of the anode sides of the cells into an anodic outlet manifold 13. The two-phase flow of oxygen and unreacted water flows from out of the cell stack 12 through the anodic outlet manifold 13.

[0024] Additionally, as noted above, in some embodiments, water may be supplied to the cathode side of the cell stack as a coolant. Accordingly, a two-phase flow of hydrogen and water is outlet from each of the cathode sides of the cells to a cathodic outlet manifold 14. The two-phase flow of hydrogen and water flows out of the cell stack 12 through the cathodic outlet manifold 14.

Electrochemical Plant

[0025] The electrochemical cells and stacks discussed within FIGS. 1A and 1B may be incorporated into an electrochemical plant having one or more electrochemical stacks (e.g., a plurality of electrochemical stacks).

[0026] The one or more electrochemical stacks may be used in the formation of a large-scale electrochemical plant that may be configured to generate at least 1,000 kg/day, at least 5,000 kg/day, or at least 10,000 kg/day of hydrogen gas using continuous operation. In certain examples, the hydrogen gas generated in the electrochemical stacks may be aggregated and supplied to an end user/customer with a purity of at least 98% at a pressure of at least 20 atm.

[0027] In other embodiments, the one or more electrochemical stacks may be used in the formation of a large-scale electrochemical plant that may be configured to generate at least 10 megawatts (MW) of power, at least 25 MW, at least 50 MW, at least 75 MW, at least 100 MW, 10-100 MW, 25-100 MW, or 50-100 MW.

[0028] The one or more electrochemical stacks may be incorporated within an electrochemical plant configured for this large-scale power generation.

[0029] FIG. 2 depicts one example of an electrochemical plant. In this particular example, the electrochemical plant is arranged or positioned in four distinct sections or segments. Specifically, the figure identifies a power supply section, an electrochemical stack section, a process equipment section, and a cooling section. The cooling section includes an embodiment of a cooling system of the present disclosure that is discussed further below. Each section may advantageously include one or more modules. Centralized piping and electrical cables may be provided within this

configuration to connect various modules to each other.

[0030] FIG. 3 depicts another example of an electrochemical plant. Similar to the example in FIG. 2, this electrochemical plant includes four distinct sections or segments: a power supply section, an electrochemical stack section, a process equipment section, and a cooling section including an embodiment of a cooling system of the present disclosure, which is discussed further below. Each section may advantageously include one or more modules.

[0031] In this particular example, the electrochemical plant includes a power supply section having four power supply modules 22 positioned in a linear arrangement and a power distribution center 24. In this particular example, each power supply module 22 includes two power supply units with centralized connections between the two units configured to supply power to a load (e.g., an electrochemical stack, a processing unit, or a cooling unit within the plant). Additional or fewer power supply units may be developed/built with each power supply module.

[0032] The power supply units within the power supply modules 22 may be connected to and receive energy from the power grid or a renewable energy power source (e.g., a solar plant, windfarm, fuel cell array). In certain examples, each power supply module and the plurality of power supply units within the power supply modules may be connected to a single input source of power. Each power supply module may be developed/manufactured off-site and transported to the plant site on a skid for quicker installation with the surrounding plant sections. In certain examples, each skid may be transported and installed onsite using pylons, concrete slabs, or screws/helical piles, which advantageously eliminate a need for large volumes of poured concrete at the plant site.

[0033] The power supply modules may further include one or more medium voltage transformers rated in a range of 1-70 kV and one or more AC-to-DC power converters. For example, the transformers may be configured to convert 6.25 MW of 34.5 kV AC to 820 V AC to feed the AC-to-DC power converters. The power converters may then transfer DC power through busbars to the electrochemical stack section.

[0034] In various implementations, the power supply modules may further include a rectifier and/or inductor to support adaptation of power from the power grid and provide power to a plurality of electrochemical stacks connected in series.

[0035] In certain examples, the power supply section of the electrochemical plant may further include a power distribution center or building 24. The power distribution center 24 may be positioned in a central location between two power supply modules in the linear arrangement of the power supply section of the plant. The power distribution center 24 may include one or more motor control centers, process logic controllers, and operator stations, wherein the power distribution center is configured to control the power distribution to the electrochemical stacks and the operation of the electrochemical stack section, process equipment section, and cooling section.

[0036] As depicted in FIG. 3, the electrochemical plant may include a plurality of electrolysis modules 26 positioned in a linear arrangement in the center of the plant for ease of deployment or for capacity additions. Fewer or more modules may be present within the plant. Further, as shown in FIG. 3, the electrolysis modules 26 may be positioned adjacent to the power supply modules 22. In this particular example, one electrolysis module is configured to be connected to and receive power from a single power supply module. Alternative arrangements are also possible wherein two power supply modules provide power to a single electrolysis module, or a single power supply module provides power to two electrolysis modules.

[0037] In this particular example, each electrolysis module 26 includes four separate electrochemical stacks. Fewer or more stacks may be present for a particular module.

[0038] In this particular example, centralized piping and electrical cables may be present within each module and between the various electrolysis modules. For example, a first electrolysis module 26 may have shared piping distributing the inlet water to the various stacks as well as shared piping for collecting/transferring the produced hydrogen and shared piping for collecting/transferring the produced oxygen from the stacks. In one example, four stacks within a single electrolysis module

26 may be connected and fed with single continuous manifold and are capable of generating at least 1,000 kg/day, at least 5,000 kg/day, or at least 10,000 kg/day of hydrogen gas using continuous operation.

[0039] Further, shared and centrally located electrical cables may be provided from a power supply module **22** adjacent to the respective stack module.

[0040] In certain examples, a minimized amount of piping may be configured to attach one electrolysis module with an additional, adjacent electrolysis module.

[0041] Returning to FIG. **3**, the electrochemical plant may further include a process equipment section or segment of the plant that may be positioned in a linear arrangement between the electrochemical stack section and the cooling section. The process equipment section, like the other sections of the plant, may include one or more processing modules.

[0042] For example, the process equipment section may include various modules such as an anode/cathode gas separation module, a hydrogen product processing module, a feed water treatment module, and/or a process water heat exchange and pumping module. Fewer or additional modules may also be included, depending on the overall size of the plant. In certain examples, the hydrogen product processing module may be developed to include a condenser, a water knockout drum and coalescing filter, allowing for high hydrogen purity to be achieved without a need for a dedicated dryer module.

[0043] As depicted in FIG. **3**, the process equipment section of the electrochemical plant may include a RO/DI unit **28** configured to receive utility water **54** and send off waste water **56**, an oxygen separator **30**, a makeup water module **32**, a makeup water tank **34**, a hydrogen separator **36**, a hydrogen cooler module **38**, an anode water pump module **40**, a cooling water pump module **42**, a cathode water pump module **44**, cathode pumps **46**, an air compressor **48**, a chiller **50**, one or more vent stacks **58**, and a nitrogen rank **62**, for example.

[0044] The process equipment section may also include process equipment modules related to feed water treatment, water circulation, and anode/cathode water level balancing. For example, a module may be associated with a RO/DI (reverse osmosis/deionization) water treatment module configured to receive utility water and treat the utility water for use within the electrochemical stacks. An additional module may be associated with the makeup water tank configured to receive the treated water from the RO/DI module and provide water to the anode and cathode gas separators for distribution to the electrochemical stacks. In some examples, the anode gas separator has a volume or capacity of at least 12,000 liters, and the cathode gas separator has a volume or capacity of at least 3,500 liters. With such capacities, the anode and cathode separators may be configured to process or accommodate an electrochemical plant configured to generate or produce at least 10,000 kg/day of hydrogen gas. In certain examples, the hydrogen gas generated in the electrochemical plant may be aggregated and supplied to an end user/customer with a purity of at least 98% at a pressure of at least 20 atm.

[0045] Additional modules within the process equipment section of the plant may be configured to provide anode water and cooling water to the electrochemical stacks and receive anode product (e.g., water and oxygen gas) from the stacks.

[0046] Returning back to FIG. **3**, the electrochemical plant further includes a cooling section or segment of the plant that may be positioned in a linear arrangement adjacent to the process equipment section. The cooling section of the plant may include an embodiment of a cooling system according to the present disclosure. The cooling system may advantageously use a single loop to cool any of the above-mentioned modules of the plant. Embodiments of the cooling system according to the present disclosure are described below with reference to FIGS. **4** and **5**.

[0047] The circled area/enclosure **64** within FIG. **3** identifies a hazardous area or section of the plant that includes these specific modules. Through this arrangement, the overall hazardous area of the plant may be minimized. For example, less than one third (33%) of the total plant area may be designated as a hazardous area, therein reducing or minimizing the cost/amount of safety

equipment required for the plant.

Cooling System For An Electrochemical Plant

[0048] FIGS. **4** and **5** depict an embodiment of a cooling system **200** for an electrochemical plant according to the present disclosure.

[0049] The cooling system **200**, as depicted in FIGS. **4** and **5**, includes a cooling module and a cooling loop. The cooling module includes a plurality of dry coolers or wet coolers configured to transfer coolant and reject waste heat generated in a plurality of separate modules of the electrochemical plant. The plurality of separate modules may be any of the above-mentioned modules. However, in these depicted examples, the plurality of separate modules includes an electrolysis module and a power supply module.

[0050] As mentioned above, the cooling system **200** also includes a cooling loop. The cooling loop includes cooling lines configured to provide fluid communication between the cooling module and each module of the plurality of separate modules within the electrochemical plant. The plurality of dry or wet coolers within the cooling modules may be connected (e.g., in parallel) to a same/singular manifold.

[0051] In certain examples, the cooling loop may also include one or more pressure regulators (not illustrated) and flow control valves (not illustrated), configured to control the coolant flowing through the cooling loop such as to balance the flow and/or prevent over pressurization in the cooling loop. The cooling loop may also include one or more sensors (not illustrated) to measure the pressure and/or temperature of the coolant flowing through the cooling lines.

[0052] In certain examples, the cooling system **200** may also include at least one pump configured to adjust (e.g., raise) a pressure of the coolant flowing through the cooling loop. In this depicted example, the cooling system **200** includes power electronics cooling booster pumps configured to raise a pressure of the coolant flowing through the cooling loop to the power supply module. The cooling system **200** also includes main cooling pumps configured to raise a pressure of the coolant flowing into the cooling module. However, any number of pumps may be used, and the present disclosure is not limited to the power electronics cooling booster pumps and the main cooling pumps.

[0053] In certain examples, the cooling system **200** may further include a controller **270** and a data acquisition unit **272**. The controller **270** is in communication with the pressure sensors, the flow control valves, the pressure regulators, the pumps, and any additional sensors within the system. The data acquisition unit **272** may be operable to measure, monitor, and/or receive system data in real-time.

[0054] For example, the controller **270** may be configured to control a pressure of the coolant flowing through the cooling lines via an adjustment to at least one pressure regulator of the cooling loop. The controller **270** may further be configured to control a flow rate of the coolant flowing through the cooling lines via an adjustment to at least one flow control valve of the cooling loop. The controller **270** may also be configured to control the pumps to pressurize the coolant in the cooling loop.

[0055] In certain examples, one or more control valves within the cooling loop may be configured to control a flow rate of the coolant entering and/or leaving the plurality of wet or dry coolers. The flow rate of the coolant entering and/or leaving the wet or dry coolers may be any configurable flow rate taking into consideration the operating parameters of the electrochemical system. In certain examples, the flow rate may be at least 0.1 liters/minute (L/min), at least 1 L/min, at least 10 L/min, at least 100 L/min, at least 200 L/min, at least 1000 L/min, at least 2000 L/min, in a range of 0.1-2000 liters per minute (L/min), 10-2000 L/min, 100-2000 L/min, 0.1-200 L/min, 1-200 L/min, 10-200 L/min, 0.1-1000 L/min, 1-1000 L/min, or 10-1000 L/min.

[0056] Furthermore, a pressure sensor may be configured to transmit the pressure reading to the data acquisition unit **272** and the data acquisition unit **272** may then transmit the information to the controller **270** so that the pressure within the cooling lines may be controlled. In other words, a

controller in communication with the pressure sensor may be configured to send a signal to either the pressure regulator or flow control valve to open the valve (partially or fully) when the pressure reading exceeds a predefined threshold level. Additionally, the controller **270** may be configured to send a signal to either the pressure regulator or flow control valve to close the valve (partially or fully) when the pressure level drops below a predefined threshold level.

[0057] The controller **270** is further configured to regulate the flow rate of the coolant within the cooling loop for the purpose of cooling the various distinct modules within the electrochemical plant. For instance, the controller **270** can receive data from sensors placed along the cooling loop. Using this data, the controller **270** determines the optimal temperature and pressure needed for effectively cooling both the plurality of separate modules, e.g., using a minimum amount of energy consumption in the cooling process. In other words, the cooling system is configured to operate at the determined optimal temperature needed for cooling both the electrolysis module and the power supply module.

[0058] Subsequently, the controller **270** fine-tunes the pressure and temperature of the coolant, using the pressure regulators and control valves, based on the identified optimal temperature and pressure before supplying the coolant to each individual module connected to the cooling loop. Moreover, the controller **270** has the capability to adjust the coolant's characteristics, utilizing pressure regulators and control valves, to ensure a consistent coolant temperature and pressure provided to each module.

[0059] In certain examples, the cooling loop is configured to provide coolant to the plurality of modules from the plurality of wet or dry coolers at a coolant temperature in a range of 20-50° C., in a range of 20-40° C., or in a range of 30-50° C. Further, the same cooling loop is configured to receive coolant from the plurality of modules at the plurality of wet or dry coolers at a coolant temperature in a range of 50-90° C., in a range of 50-80° C., in a range of 50-70° C., in a range of 60-90° C., in a range of 60-80° C., or in a range of 70-90° C. Through the wet or dry coolers, the coolant temperature is configured to be lowered to a desired temperature to effectively cool the plurality of modules in the next cycle or circulation of coolant. As such, the coolant temperature exiting from the plurality of wet or dry coolers needs to be low enough to provide the desired heat exchange with the plurality of modules but should not be too low that unnecessary or excess energy is being used within the plurality of wet or dry coolers to facilitate the temperature decrease of the coolant. Therefore, a maximum or optimal target temperature for the coolant leaving the wet or dry coolers should be identified that can provide effective cooling while minimizing energy/operating costs.

[0060] For example, if the electrolysis module operates most efficiently at a maximum temperature of 50° C. and a pressure of 2 atm to effectively cool the module, while the power supply module requires a maximum temperature of 40° C. and a pressure of 1.5 atm to effectively cool the module, the controller **270** may identify these as the critical parameters and determine the optimal temperature/pressure to supply coolant to both modules is at a maximum temperature of 40° C. and a pressure of 1.5 atm, such that both modules are effectively cooled.

[0061] The controller **270** then dynamically adjusts the pressure and temperature of the coolant in the cooling loop to meet these specific requirements before distributing the coolant to each module. The controller thus provides that the electrolysis module receives coolant at 40° C. and 1.5 atm, and the power supply module also receives coolant at 40° C. and 1.5 atm.

[0062] Additionally, the controller **270** utilizes pressure regulators and control valves to fine-tune the characteristics of the coolant throughout the process, therein providing a similar or set temperature and pressure to each of the various modules for optimal performance of both the electrolysis and power supply modules.

[0063] This dynamic controller helps maximize the efficiency of the electrochemical plant by tailoring the coolant conditions based on real-time data and specific operational requirements of each module.

[0064] In certain examples, the cooling system **200** may also include an expansion tank in fluid connection with the cooling loops. The expansion tank is configured to act as a buffer to absorb the excess pressure during expansion and provide additional fluid during contraction. In this depicted example, only one expansion tank is provided, however any number of expansion tanks may be used in the cooling system **200**.

[0065] Referring back to FIGS. **4** and **5**, the cooling module is configured to transfer the coolant via the coolant loop to each module of the plurality of separate modules. Additionally, the cooling module is configured to receive, via the cooling loop, the coolant from each module of the plurality of separate modules and reject the waste heat collected by the coolant received from each module of the plurality of separate modules to a surrounding environment.

[0066] In this depicted example, the cooling loop is configured to be in fluid communication with the electrolysis module. The electrolysis module may include one or more electrochemical stacks (e.g., a plurality of electrochemical stacks). The plurality of electrochemical stacks is connected via a same cathode heat exchanger and a same anode heat exchanger that are configured to be in fluid communication with the cooling module and the power supply module via the cooling lines. Additionally, the plurality of electrochemical stacks is connected via a same anode inlet water supply and a same anode outlet water supply.

[0067] Additionally, in this depicted example, the cooling loop is configured to be in fluid communication with the power supply module. The power supply module may include one or more medium voltage transformers rated in a range of 1-70 kV, one or more AC-to-DC power converters, and one or more rectifiers configured to provide power to the electrolysis module of the electrochemical plant.

[0068] Referring to FIG. **5**, the plurality of dry or wet coolers within the cooling modules may be connected (e.g., in parallel) to a same/singular manifold. A cooling line from the singular manifold is configured to transfer coolant to the anode heat exchanger, and the cathode heat exchanger (i.e., the balance of power (BOP)), which are connected to the plurality of electrochemical stacks. The cooling line from the singular manifold is also configured to transfer coolant to the power electronics cooling lines. The coolant is then received through the BOP cooling water return and transferred along the cooling line to the singular manifold connecting the plurality of dry or wet coolers within the cooling module. As a result, the cooling system **200** advantageously employs a singular cooling loop to cool multiple modules within an electrochemical plant using the same coolant at comparable temperature and pressure, thereby streamlining the overall cooling system complexity.

[0069] FIG. **6** depicts a flowchart describing a method for cooling a plurality of separate modules of the electrochemical plant. In act **S101**, the cooling module is configured to transfer coolant along a cooling loop to the plurality of separate modules of the electrochemical plant. Specifically, the dry coolers or wet coolers of the cooling module are configured to transfer the coolant via the cooling loop to each module of the plurality of separate modules.

[0070] In act **S103**, the controller determines an optimal temperature and pressure required to effectively cool both the electrolysis module and the power supply module. This may refer to a highest coolant temperature provided to each of the modules that can effectively cool each module, therein providing the minimum power consumption cost in the operation of the wet or dry coolers for the cooling process. For example, the controller identifies the optimal cooling conditions that meet the specific requirements of both modules within the electrochemical plant by first measuring the pressures and temperatures at each separate module and comparing the pressure and temperatures at each module to each other, and, e.g., identifying the operating temperature and pressure required for a minimum amount of energy consumption in the cooling of the plurality of modules within the electrochemical plant.

[0071] In act **S105**, the controller causes the cooling module to provide the coolant to both the electrolysis module and the power supply module at the determined optimal temperature and

pressure. In other words, the controller controls the plurality of dry coolers or wet coolers of the cooling module to provide the coolant to both the electrolysis module and the power supply module at the determined optimal temperature and pressure, e.g., using a minimum amount of energy consumption in the cooling process.

[0072] In act **S107**, the controller may, optionally, cause at least one pump to raise a pressure of the coolant flowing through the cooling lines.

[0073] In act **S109**, the dry cooler or wet coolers of the cooling module receive the coolant via the cooling loop from both the electrolysis module and the power supply module.

[0074] In act **S111**, the dry coolers or wet coolers of the cooling module reject the waste heat collected by the coolant received from each module of the plurality of separate modules to a surrounding environment.

Controlling Operation of the Cooling System

[0075] FIG. 7 illustrates an exemplary system **120** for controlling operation of a cooling system **200** for an electrochemical plant. The system **120** includes the cooling system **200** (such as depicted in FIGS. 4 and 5), a monitoring system **121**, a workstation **128**, and a network **127**. Additional, different, or fewer components may be provided.

[0076] The monitoring system **121** includes a server **125** and a database **123**. The monitoring system **121** may include computer systems and networks of a system operator (e.g., the operator of the cooling system **200**). The server database **123** may be configured to store information regarding the operating conditions or setpoints for optimizing the performance of the cooling system **200**.

[0077] The monitoring system **121**, the workstation **128**, and the cooling system **200** are coupled with the network **127**. The phrase “coupled with” is defined to mean directly connected to or indirectly connected through one or more intermediate components. Such intermediate components may include hardware and/or software-based components.

[0078] The optional workstation **128** may be a general-purpose computer including programming specialized for providing input to the server **125**. For example, the workstation **128** may provide settings for the server **125**. The workstation **128** may include at least a memory, a processor, and a communication interface.

[0079] FIG. 8 illustrates an exemplary server **125** of the system of FIG. 7. The server **125** includes a memory **274**, a controller or processor **270**, and a communication interface **276**. The server **125** may be coupled to a database **123** and a workstation **128**. The workstation **128** may be used as an input device for the server **125**. The communication interface **276** receives data indicative of use inputs made via the workstation **128** or a separate electronic device.

[0080] The controller or processor **270** may include a general processor, digital signal processor, an application specific integrated circuit (ASIC), field programmable gate array (FPGA), analog circuit, digital circuit, combinations thereof, or other now known or later developed processor. The controller or processor **270** may be a single device or combination of devices, such as associated with a network, distributed processing, or cloud computing that is configured to control operation of one or more components of the cooling system **200**.

[0081] The memory **274** may be a volatile memory or a non-volatile memory. The memory **274** may include one or more of a read only memory (ROM), random access memory (RAM), a flash memory, an electronic erasable program read only memory (EEPROM), or other type of memory. The memory **274** may be removable from the device **122**, such as a secure digital (SD) memory card.

[0082] The communication interface **276** may include any operable connection. An operable connection may be one in which signals, physical communications, and/or logical communications may be sent and/or received. An operable connection may include a physical interface, an electrical interface, and/or a data interface. The communication interface **276** provides for wireless and/or wired communications in any now known or later developed format.

[0083] In the above-described examples, the network **127** may include wired networks, wireless

networks, or combinations thereof. The wireless network may be a cellular telephone network, an 802.11, 802.16, 802.20, or WiMax network. Further, the network 127 may be a public network, such as the Internet, a private network, such as an intranet, or combinations thereof, and may utilize a variety of networking protocols now available or later developed including, but not limited to TCP/IP based networking protocols.

[0084] While the non-transitory computer-readable medium is described to be a single medium, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

[0085] In a particular non-limiting example, the computer-readable medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium can be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium can include a magneto-optical or optical medium, such as a disk or tapes or other storage device to capture carrier wave signals such as a signal communicated over a transmission medium. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

[0086] In an alternative example, dedicated hardware implementations, such as application specific integrated circuits, programmable logic arrays and other hardware devices, can be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various examples can broadly include a variety of electronic and computer systems. One or more examples described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that can be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

[0087] In accordance with various embodiments of the present disclosure, the methods described herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limited embodiment, implementations can include distributed processing, component/object distributed processing, and parallel processing. Alternatively, virtual computer system processing can be constructed to implement one or more of the methods or functionalities as described herein.

[0088] Although the present specification describes components and functions that may be implemented in particular embodiments with reference to particular standards and protocols, the claim scope is not limited to such standards and protocols. For example, standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP, HTTPS) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having similar functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.

[0089] A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a standalone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup

language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

[0090] The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

[0091] As used in this application, the term “circuitry” or “circuit” refers to all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry) and (b) to combinations of circuits and software (and/or firmware), such as (as applicable): (i) to a combination of processor(s) or (ii) to portions of processor(s)/software (including digital signal processor(s)), software, and memory (ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions) and (c) to circuits, such as a microprocessor(s) or a portion of a microprocessor(s), that require software or firmware for operation, even if the software or firmware is not physically present.

[0092] This definition of “circuitry” applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term “circuitry” would also cover an implementation of merely a processor (or multiple processors) or portion of a processor and its (or their) accompanying software and/or firmware. The term “circuitry” would also cover, for example and if applicable to the particular claim element, a baseband integrated circuit or applications processor integrated circuit for a mobile phone or a similar integrated circuit in server, a cellular network device, or other network device.

[0093] Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and anyone or more processors of any digital computer. The processor may receive instructions and data from a read only memory or a random-access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. The computer may also include or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a Global Positioning System (GPS) receiver, to name just a few. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media, and memory devices, including by way of example semiconductor memory devices, e.g., E PROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0094] To provide for interaction with a user, embodiments of the subject matter described in this specification can be implemented on a device having a display, e.g., a CRT (cathode ray tube), LCD (liquid crystal display), or LED (light emitting diode) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input.

[0095] Embodiments of the subject matter described in this specification can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a

middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described in this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), e.g., the Internet.

[0096] The computing system can include clients and servers. A client and server may be remote from each other and may interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship with each other.

[0097] One or more embodiments of the disclosure may be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any particular invention or inventive concept. Moreover, although specific embodiments have been illustrated and described herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, are apparent to those of skill in the art upon reviewing the description.

[0098] As used herein, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise.

[0099] As used herein, “for example,” “for instance,” “such as,” or “including” are meant to introduce examples that further clarify more general subject matter. Unless otherwise expressly indicated, such examples are provided only as an aid for understanding embodiments illustrated in the present disclosure and are not meant to be limiting in any fashion. Nor do these phrases indicate any kind of preference for the disclosed embodiment.

[0100] The Abstract of the Disclosure is provided to comply with 37 C.F.R. § 1.72 (b) and is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure. This disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may be directed to less than all of the features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

[0101] It is intended that the foregoing detailed description be regarded as illustrative rather than limiting and that it is understood that the following claims including all equivalents are intended to define the scope of the disclosure. The claims should not be read as limited to the described order or elements unless stated to that effect. Therefore, all embodiments that come within the scope and spirit of the following claims and equivalents thereto are claimed as the disclosure.

Claims

1. A cooling system for an electrochemical plant, the cooling system comprising: a cooling module including a plurality of dry coolers or wet coolers configured to transfer a coolant and reject waste heat generated in a plurality of separate modules of the electrochemical plant; and a cooling loop including cooling lines configured to provide fluid communication between the cooling module and each module of the plurality of separate modules within the electrochemical plant, wherein the cooling module is configured to transfer the coolant via the cooling loop to each module of the

plurality of separate modules, and wherein the cooling module is configured to receive, via the cooling loop, the coolant from each module of the plurality of separate modules and reject the waste heat collected by the coolant received from each module of the plurality of separate modules to a surrounding environment.

2. The cooling system of claim 1, further comprising: a controller configured to determine an optimal temperature and pressure required to cool each module of the plurality of separate modules at a minimum operating cost for the electrochemical plant.
3. The cooling system of claim 2, wherein the controller is further configured to provide the coolant to each module of the plurality of separate modules at the determined temperature and pressure.
4. The cooling system of claim 2, wherein the controller is further configured to provide the coolant at a set temperature and pressure to each module of the plurality of separate modules.
5. The cooling system of claim 2, wherein the controller is further configured to: control a pressure of the coolant flowing through the cooling lines via an adjustment to at least one pressure regulator of the cooling loop; and control a flow rate of the coolant flowing through the cooling lines via an adjustment to at least one flow control valve of the cooling loop.
6. The cooling system of claim 2, wherein the controller is further configured to raise a pressure of the coolant flowing through the cooling lines via an adjustment to at least one pump positioned within the cooling loop.
7. The cooling system of claim 1, wherein the plurality of separate modules comprises an electrolysis module and a power supply module, wherein the electrolysis module comprises a plurality of electrochemical stacks connected via a same cathode heat exchanger and a same anode heat exchanger and configured to be in fluid communication with the cooling module and the power supply module via the cooling lines, and wherein the plurality of electrochemical stacks is connected via a same anode inlet water supply and a same anode outlet water supply.
8. The cooling system of claim 7, wherein each electrochemical stack of the plurality of electrochemical stacks comprises a plurality of electrochemical cells, and wherein each electrochemical cell within a respective electrochemical stack is configured to operate with 200 mV or less of pure resistive loss when operating at a current density of at least 3 Amps/cm².
9. The cooling system of claim 7, wherein the power supply module comprises one or more medium voltage transformers rated in a range of 1 to 70 kV, one or more AC-to-DC power converters, and one or more rectifiers configured to provide power to the electrolysis module of the electrochemical plant.
10. The cooling system of claim 1, wherein the cooling loop further comprises one or more pressure regulators, one or more flow control valves, or a combination of one or more pressure regulators and one or more flow control valves configured to control the coolant flowing through the cooling loop.
11. The cooling system of claim 1, wherein a flow rate of the coolant within the cooling loop being received by the cooling module and/or being transferred to the plurality of separate modules is configured to be in a range of 0.1-2000 liters per minute.
12. The cooling system of claim 1, further comprising: at least one pump configured to raise a pressure of the coolant flowing through the cooling loop.
13. The cooling system of claim 1, wherein a temperature of the coolant being transferred from the cooling module to the plurality of separate modules is configured to be in a range of 20-50° C., and wherein a temperature of the coolant being received by the cooling module from the plurality of separate modules is configured to be in a range of 50-90° C.
14. A method for cooling an electrochemical plant comprising a plurality of separate modules using a cooling system, the method comprising: transferring a coolant via a cooling loop of the cooling system, by a cooling module of the cooling system, to each module of the plurality of separate modules of the electrochemical plant; receiving coolant, via the cooling loop of the cooling system, from each module of the plurality of separate modules; and rejecting, by the cooling module, waste

heat collected by the coolant received from each module of the plurality of separate modules to a surrounding environment, wherein the cooling module comprises a plurality of dry coolers or a plurality of wet coolers, and wherein the cooling loop comprises cooling lines that provide fluid communication between the cooling module and each module of the plurality of separate modules within the electrochemical plant.

15. The method of claim 14, further comprising: determining, by a controller of the cooling system, an optimal temperature and pressure required to cool each module of the plurality of separate modules at a minimum operating cost for the electrochemical plant.

16. The method of claim 15, further comprising: controlling, by the controller, the plurality of dry coolers or wet coolers to provide the coolant to each module of the plurality of separate modules at the determined temperature and pressure.

17. The method of claim 15, wherein the cooling loop further comprises one or more pressure regulators, one or more flow control valves, or a combination of one or more pressure regulators and one or more flow control valves configured to control the coolant flowing through the cooling loop, and wherein the method further comprises controlling, by the controller, the one or more pressure regulators, the one or more flow control valves, or a combination thereof, to provide the coolant at a set temperature and pressure to each module of the plurality of separate modules.

18. The method of claim 17, further comprising: controlling, by the controller, a pressure of the coolant flowing through the cooling lines via an adjustment to at least one pressure regulator of the cooling loop; and/or controlling, by the controller, a flow rate of the coolant flowing through the cooling lines via an adjustment to at least one flow control valve of the cooling loop.

19. The method of claim 15, further comprising: raising, by the controller, a pressure of the coolant flowing through the cooling lines via an adjustment to at least one pump.

20. The method of claim 14, wherein the plurality of separate modules comprises an electrolysis module and a power supply module, wherein the electrolysis module comprises a plurality of electrochemical stacks that are connected via a same cathode heat exchanger and a same anode heat exchanger configured to be in fluid communication with the cooling module and the power supply module via the cooling lines, and wherein the plurality of electrochemical stacks is connected via a same anode inlet water supply and a same anode outlet water supply.
